EIN 4343 Inventory and Supply Chain Systems
Spring 26: Homework Assignment 6

1. **[2 points] Service Levels Revisited:** Fill in the blanks with the appropriate parameter using one of the following:
   - **a.** The fraction of demand that can<u>not</u> be satisfied from inventory directly is denoted by $1-\beta$
   - **b.** The probability of stocking out during lead time is denoted by $1-\alpha$
   - **c.** The probability of <u>not</u> experiencing a stock out during lead time is denoted by α
   - **d.** The fraction of demand that can be satisfied from inventory directly is denoted by β

2. **[7 points] (2016 Final Exam Question)** AW Inc. produces two variants of a particular product, P1 and P2, using two types of components, C1 and C2. P1 requires 1 unit of C1 and 2 units of C2, and is assembled in Seattle at a rate of 60 units/day. P2 requires 3 units of C1 and 1 unit of C2, and is assembled in Redding at a rate of 80 units/day. C1 and C2 are manufactured by suppliers in Chicago and Atlanta, respectively. The assembly facilities operate 360 days/year. AW Inc. outsources transportation operations to a third-party logistics (TPL) provider. The TPL provider offers trucks that can carry 3,000 units of either C1 or C2 at a cost of $3,200 per trip. The delivery lead-time from Atlanta to either Seattle or Redding is 5 days, whereas the delivery lead time from Chicago to either Seattle or Redding is 4 days. AW Inc. incurs an inventory holding cost of $0.35/unit/day for components that are stocked at the assembly plants and $0.25/unit/day for components that are in-transit from the supplier locations to the assembly plants.
   - **a. [3 points]** What are the total annual logistics costs if full truckload direct deliveries are made from the supplier plants to the assembly plants?

Transportation cost: C1: Chi→ Sea	$(3{,}200)\dfrac{(1)(60)(360)}{(3{,}000)} = \$23{,}040/\text{year}$
Transportation cost: C2: Atl→ Sea	$(3{,}200)\dfrac{(2)(60)(360)}{(3{,}000)} = \$46{,}080/\text{year}$
Transportation cost: C1: Chi→ Red	$(3{,}200)\dfrac{(3)(80)(360)}{(3{,}000)} = \$92{,}160/\text{year}$
Transportation cost: C2: A**tl**→ Red	$(3{,}200)\dfrac{(1)(80)(360)}{(3{,}000)} = \$30{,}720/\text{year}$

In-trans inv hld cost: C1: Chi→ Sea	$(0.25)(360)(1)(60)(360)\frac{4}{(360)} = \$21{,}600/\text{year}$
In-trans inv hld cost: C2: Atl→ Sea	$(0.25)(360)(2)(60)(360)\frac{4}{(360)} = \$43{,}200/\text{year}$
In-trans inv hld cost: C1: Chi→ Red	$(0.25)(360)(3)(80)(360)\frac{4}{(360)} = \$86{,}400/\text{year}$
In-trans inv hld cost: C2: Atl→Red	$(0.25)(360)(1)(80)(360)\frac{4}{(360)} = \$28{,}800/\text{year}$

On-site inv hld cost: C1 @ Sea	$(0.35)(360)\frac{3{,}000}{2} = \$189{,}000/\text{year}$
On-site inv hld cost: C2 @ Sea	$(0.35)(360)\frac{3{,}000}{2} = \$189{,}000/\text{year}$
On-site inv hld cost: C1 @ Red	$(0.35)(360)\frac{3{,}000}{2} = \$189{,}000/\text{year}$
On-site inv hld cost: C2 @ Red	$(0.35)(360)\frac{3{,}000}{2} = \$189{,}000/\text{year}$

b. Self study: You do not need to solve this and submit as part of the homework assignment. However, solve this on your own time as exercise. What are the total annual logistics costs if optimal shipment sizes were to be used for direct deliveries?

Opt Shipment Size: C1: Chi→Sea	$\sqrt{\frac{2(3{,}200)(1)(60)(360)}{(0.35)(360)}} = 1{,}047.45 \text{ units}$
Opt Shipment Size:C2: Atl→Sea	$\sqrt{\frac{2(3{,}200)(2)(60)(360)}{(0.35)(360)}} = 1{,}481.31 \text{ units}$
Opt Shipment Size:C1: Chi→Red	$\sqrt{\frac{2(3{,}200)(3)(80)(360)}{(0.35)(360)}} = 2{,}094.89 \text{ units}$
Opt Shipment Size: C2: Atl→Red	$\sqrt{\frac{2(3{,}200)(1)(80)(360)}{(0.35)(360)}} = 1{,}209.49 \text{ units}$

Transportation cost: C1: Chi→ Sea	$(3{,}200)\frac{(1)(60)(360)}{(1{,}047.45)} = \$65{,}989.09/\text{year}$
Transportation cost: C2: Atl→Sea	$(3{,}200)\frac{(2)(60)(360)}{(1{,}481.31)} = \$93{,}322.67/\text{year}$
Transportation cost: C1: Chi→Red	$(3{,}200)\frac{(3)(80)(360)}{(2{,}094.89)} = \$131{,}978.18/\text{year}$
Transportation cost: C2: A**tl**→Red	$(3{,}200)\frac{(1)(80)(360)}{(1{,}209.49)} = \$76{,}197.64/\text{year}$

In-trans inv hld cost: C1: Chi→ Sea	$(0.25)(360)(1)(60)(360)\frac{4}{(360)} = \$21{,}600/\text{year}$
In-trans inv hld cost: C2: Atl→ Sea	$(0.25)(360)(2)(60)(360)\frac{4}{(360)} = \$43{,}200/\text{year}$
In-trans inv hld cost: C1: Chi→ Red	$(0.25)(360)(3)(80)(360)\frac{4}{(360)} = \$86{,}400/\text{year}$
In-trans inv hld cost: C2: Atl→Red	$(0.25)(360)(1)(80)(360)\frac{4}{(360)} = \$28{,}800/\text{year}$

On-site inv hld cost: C1 @ Sea	$(0.35)(360)\frac{3{,}000}{2} = \$65{,}989.09/\text{year}$
On-site inv hld cost: C2 @ Sea	$(0.35)(360)\frac{3{,}000}{2} = \$93{,}322.67/\text{year}$
On-site inv hld cost: C1 @ Red	$(0.35)(360)\frac{3{,}000}{2} = \$131{,}978.18/\text{year}$
On-site inv hld cost: C2 @ Red	$(0.35)(360)\frac{3{,}000}{2} = \$76{,}197.64/\text{year}$

c. [4 points] Suppose that the TPL provider agrees to make milk-run pick-ups and deliveries and carry less than full truckloads if necessary. AW Inc. considers a milk-run pickup scheme using optimal shipment sizes as follows: A truck would go to Chicago first to pick up C1, to Atlanta next to pick up C2, and deliver C1 and C2 to Redding. Similarly, another truck would go to Atlanta first to pick up C2, to Chicago next to pick up C1, and deliver C1 and C2 to Seattle. What would the average annual total logistics (including on-site and in-transit inventory holding along with transportation) cost incurred by AW Inc.? Suppose that the lead-time between Atlanta and Chicago is 1 day in either direction.

Opt Shipment Size: Chi→Atl → Red	$\sqrt{\frac{2(3{,}200)(3+1)(80)(360)}{(0.35)(360)}} = 2{,}418.97 \text{ units}$
Opt Shipment Size: Atl-→Chi → Sea	$\sqrt{\frac{2(3{,}200)(1+2)(60)(360)}{(0.35)(360)}} = 1{,}814.23\text{units}$

Transportation cost: C1: Chi→ Atl → Red	$(3{,}200)\frac{(3+1)(80)(360)}{(2{,}418.97)} = \$152{,}395.28/\text{year}$
Transportation cost: C2: Atl → Chi → Sea	$(3{,}200)\frac{(1+2)(60)(360)}{(1{,}814.23)} = \$114{,}296.46/\text{year}$

In-trans inv hld cost: C1: Chi→ Atl→Red	$(0.25)(360)\left[(3)(80)(360)\frac{1+5}{(360)}+(1)(80)(360)\frac{5}{(360)}\right]$ $= 129{,}600 + 36{,}000 = \$165.600/\text{year}$
In-trans inv hld cost: C2: Atl→Chi→Red	$(0.25)(360)\left[(2)(60)(360)\frac{1+4}{(360)}+(1)(60)(360)\frac{4}{(360)}\right]$ $= 54{,}000 + 21{,}600 = \$75{,}600/\text{year}$

On-site inv hld cost: Chi→ Atl→ Red	$(0.35)(360)\frac{2{,}418.97}{2} = \$152{,}395.28/\text{year}$
On-site inv hld cost: Atl→ Chi→ Sea	$(0.35)(360)\frac{1{,}814.23}{2} = \$114{,}296.46/\text{year}$